

# Application of retention basins and on-contour swales for flood risk reduction

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RECEIVED 27.11.2025

ACCEPTED 27.03.2026

AVAILABLE ONLINE 17.09.2026

**Abstract:** This study evaluates the effectiveness of non-structural watershed management measures for flood risk reduction, with a focus on the Mirusha River catchment and the urban area of Malisheva. The research investigates the application of retention ponds integrated with contour-aligned channels (swales), following the principles of Keyline Design, as a strategy to control surface runoff, enhance infiltration, and mitigate urban flooding.

The methodology combines long-term climatic and hydrological data analysis, runoff estimation using multiple hydrological methods, and GIS-based spatial analysis to identify optimal locations for retention ponds. Sixteen retention ponds were strategically positioned at keypoints along contour lines to maximise water retention and runoff attenuation.

The results demonstrate that the proposed system is capable of retaining approximately 903,840.67 m<sup>3</sup> of water annually, which corresponds to a reduction of the flood-prone area by approximately 180.77 ha under a 50-year return period flood scenario. Beyond flood mitigation, the system contributes to increased groundwater recharge, reduced peak flows, and enhanced hydrological resilience at the catchment scale.

The findings confirm that the integration of retention ponds and on-contour swales represents an effective and sustainable non-structural approach to flood risk management, offering simultaneous hydrological, ecological, and socio-economic benefits for vulnerable urban and peri-urban areas.

**Keywords:** catchment management, contour swales, flood mitigation, non-structural measures, retention ponds

## INTRODUCTION

In recent decades, the increase in extreme flood events has posed significant challenges for water management. Climate change and urban expansion present new challenges for the management of urban stormwater, making the preservation of aquatic ecosystems increasingly sensitive (Abduljaleel *et al.*, 2023). Structural measures, such as dams and large channels, are often expensive and unsuitable for urban or peri-urban areas, highlighting the need for non-structural measures, which utilise natural processes and topographic configuration to reduce runoff and enhance water infiltration.

Non-structural measures for water management represent a sustainable approach that does not require major engineering interventions but leverages natural processes and the topographic

configuration of the terrain. These measures aim to slow surface runoff and attenuate flood waves, increasing groundwater infiltration and improving hydrological balance across catchments. Contemporary literature has documented that non-structural methods, including retention ponds placed along contour lines, can contribute to water quality improvement and the protection of aquatic ecosystems, demonstrating substantial potential in water management.

Within the framework of non-structural flood mitigation measures, a representative example in Europe is the implementation of retention basins along the Lura River in the Province of Como, Italy. This project has been developed as a nature-based solution for surface water management and flood risk reduction in a densely built urban–peri-urban area experiencing intense rainfall (Oppla, 2017). The system consists of two interconnected

basins and an open channel, which are flooded in a controlled manner during intense rain events, allowing the capture and delay of peak flows. The total retention capacity of approximately 340,000 m<sup>3</sup> significantly contributes to reducing rapid runoff towards urban areas and mitigating flooding in the lower part of the valley.

Another study presents an empirical analysis of a retention basin in an agricultural field in southern Ireland, where the structure was designed as a nature-based measure to reduce peak flows and protect surrounding water systems. During a two-year monitoring period of intense rainfall events, peak flow reduction reached less than 38% compared to flows without intervention (Murphy *et al.*, 2025).

Additionally, within the framework of non-structural flood mitigation measures, Keyline Design represents an innovative approach for surface water management and enhancement of soil retention capacity. Developed by P.A. Yeomans in Australia in the mid-20th century, this system aims to slow surface runoff, distribute water across slopes, and increase infiltration, thereby reducing erosion and minimising flood risk (Yeomans, 1954). At the plains property in Queensland, Australia, channels oriented along contour lines were constructed to distribute surface water towards a main reservoir. This practice allows the delay of peak flows, increases infiltration, and reduces water concentrations that could trigger flooding in lower areas, while simultaneously maximising water capture for agricultural use (Yeomans, 1954; Yeomans, 1958).

In modern mountain farm restoration projects in Queensland, channels and swales were placed precisely along main contour lines, slowing surface runoff, increasing infiltration, and restoring vegetation. Monitoring indicated that this approach reduces the volume of concentrated runoff and flood risk, demonstrating the potential of Keyline as a non-structural strategy for surface water management (Giambastiani *et al.*, 2023).

However, non-structural measures are not always sufficient to manage extreme flooding. Their effectiveness can be influenced

by terrain characteristics, land typology, as well as their placement and design. These structures require regular maintenance to preserve functionality, and failure to adhere to such practices can significantly reduce their hydrological performance. In cases of heavy rainfall, retention ponds and contour channels may still be insufficient to prevent flooding.

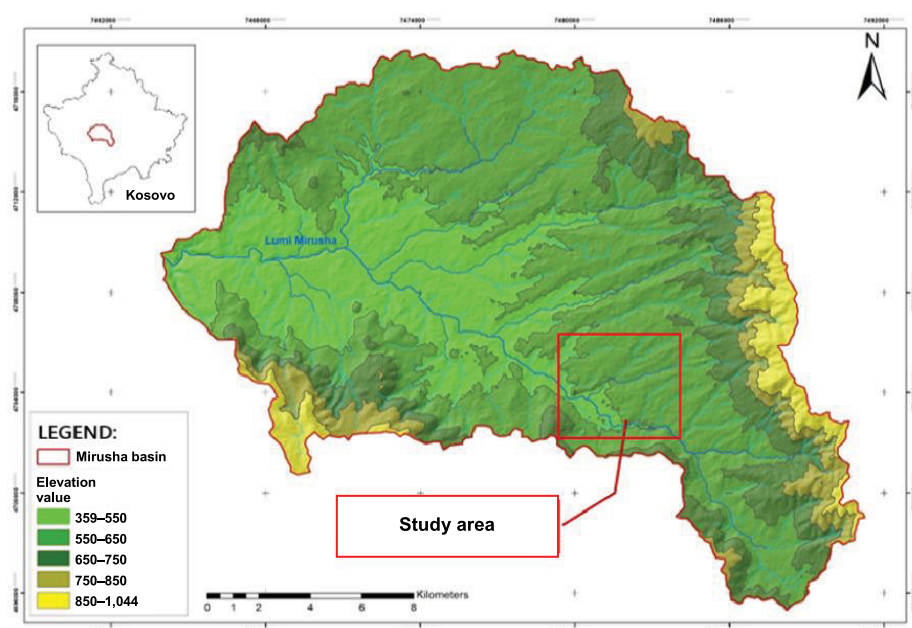
This study aims to assess the effectiveness of non-structural measures, specifically ponds integrated with contour channels, in reducing flood risk in the Mirusha River catchment and the urban area of Malisheva. The study provides original insights into the quantitative evaluation of flood reduction potential, opportunities for agricultural irrigation, and groundwater level enhancement, proposing a locally adapted and sustainable strategy for water management.

## STUDY MATERIALS AND METHODS

### STUDY AREA

The Mirusha River is located in the eastern part of the Dukagjini Plain (Ibrahimi *et al.*, 2024) and represents a sub-catchment of the White Drin River basin, extending primarily within the territory of the Municipality of Malisheva (Fig. 1). The Mirusha River catchment covers an area of approximately 335 km<sup>2</sup> and has a length of 38 km from its source to its confluence with the main river (AMMK, 2020; Ibrahimi *et al.*, 2024). The climate of the area is Mediterranean–continental, with annual precipitation ranging from 650 to 800 mm.

The study area includes a part of the urban and peri-urban zones of the city of Malisheva, as well as several rural settlements located along the Mirusha River (Fig. 2). This area is exposed to flood risk during periods of intense rainfall. The dominant land use is agricultural, with fields and meadows occupying the majority of the area, contributing to increased surface runoff during heavy precipitation events. Vegetation is primarily composed of grasses and trees, which, in the absence of proper



**Fig. 1.** Location map of the study area in Malisheva Municipality, Kosovo, showing the Mirusha River; source: own elaboration

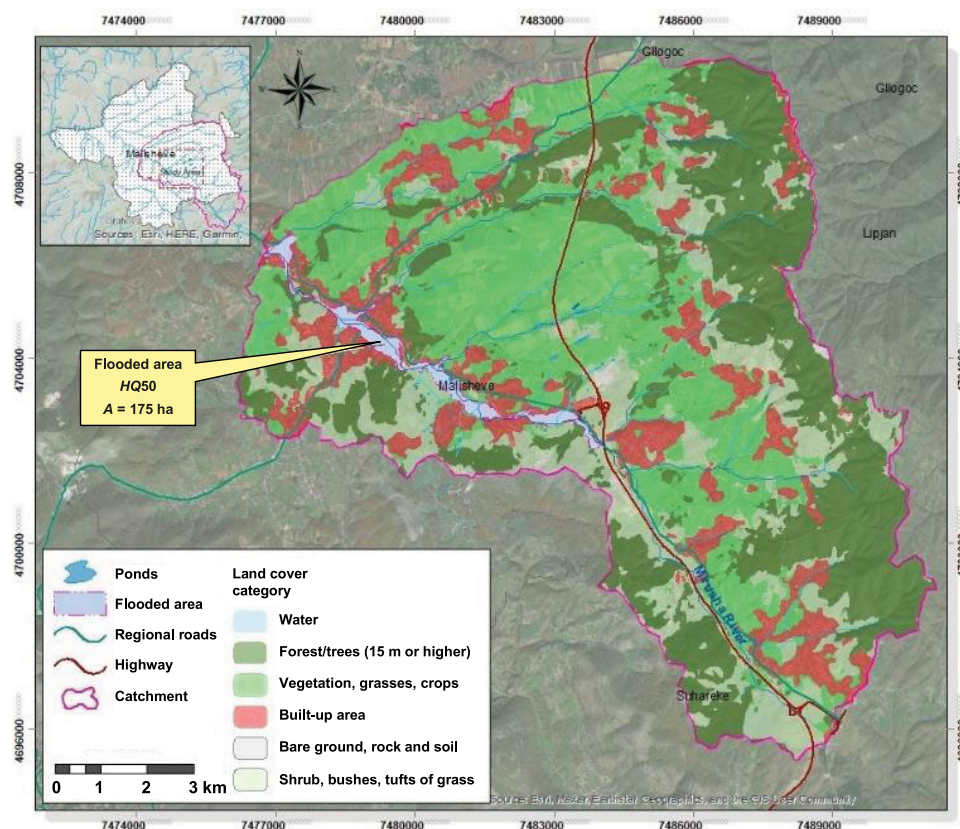


Fig. 2. Distribution of urban, agricultural, and vegetated areas within the study area; source: own elaboration

maintenance, negatively affect the river's transport capacity. Additionally, in some degraded segments, vegetation cover is limited or fragmented, increasing vulnerability to runoff, intensifying soil erosion, and worsening riverbed hydraulic conditions.

In Kosovo, repeated flood events have been recorded over the last decade (2016, 2021 and 2022) (MMPHI, 2023), primarily due to intense rainfall and the limited capacity of surface water management systems, particularly in the Dukagjini Plain (Idrizi, Nimani and Pashova, 2023).

In this context, the Mirusha River, including the territory of the Municipality of Malisheva, is continuously exposed to seasonal flooding, particularly during intense winter rainfall, when increased discharges cause riverbank overflow and flooding of surrounding areas and agricultural lands. Due to urban development along the Mirusha River, interventions on traditional structures for controlling major floods are limited. For this reason, the implementation of non-structural measures in the upper part of the catchment is considered a suitable alternative and represents the main objective of this study.

## STUDY METHODS

Climatic and hydrological data, including annual precipitation for the period 2006–2023, were obtained from the Malisheva meteorological station. For terrain and land-use analysis, 1:25,000-scale topographic maps, geological maps, and 2018 digital orthophotos were used to evaluate slope, soil characteristics, and vegetation cover. For runoff calculation, three main methods were applied: the Regrarians formula, the specific runoff isohyet method, and the runoff coefficient method (Doherty,

Jeeves and Pavlov, 2025). For each land-use class – urban areas, agricultural land, and forests – peak flows, annual water volumes, and potential flood-prone areas were determined. This approach allows a detailed understanding of surface runoff distribution at the catchment scale and the identification of areas most sensitive to flooding.

Spatial analysis was performed using a geographic information system (GIS), applying the principles of Keyline Design to identify “keypoints” along contour lines (Doherty and Jeeves, 2015). Based on Keyline Design principles, the optimal locations of 16 retention ponds in the upper part of the catchment were determined. The design of the ponds was based on suitable capacity and size for water retention, ensuring that they could delay peak flows, enhance groundwater infiltration, and provide potential irrigation for agricultural areas.

The retention of this water volume contributes to the reduction of flooded areas, improves infiltration, and provides an integrated and sustainable approach for non-structural flood management in the Mirusha River catchment. This method combines climatic and hydrological analysis with the assessment of non-structural measures, creating a robust basis for flood mitigation planning.

## Precipitation

Figure 3 presents annual precipitation for the period 2006–2023. Precipitation data were obtained from the Malisheva meteorological station. These data provide a reliable basis for analysing the temporal and spatial patterns of precipitation within the study area, which are essential for assessing runoff generation and for the design of retention ponds (IHMK, 2025).

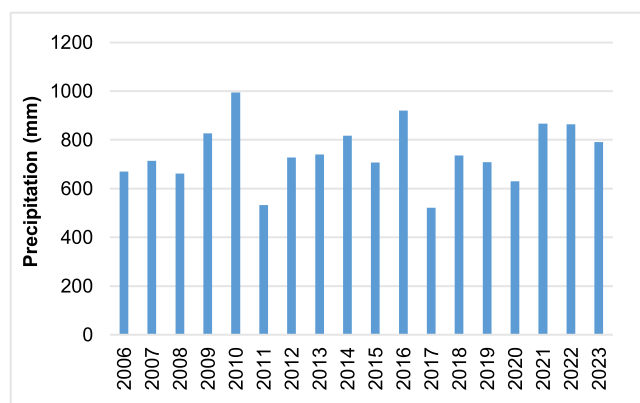


Fig. 3. Annual precipitation in the years 2006–2023 at the Malisheva meteorological station; source: own study based on data from IHMK (2025)

### Analysis of runoff data

For runoff calculation, three methods were used: the Regrarians formula (method 1), Regrarians formula – using the runoff coefficient method (method 2) and the specific runoff isohyet method (method 3).

**Method 1.** Regrarians formula (Doherty, Jeeves and Pavlov, 2025):

$$Q = 100A \cdot R \cdot Y \quad (1)$$

Runoff coefficients, expressed as a percentage of annual precipitation, were determined based on relevant literature data. For an average annual precipitation of 746.21 mm, with a total evaporation of less than 1300 mm and a climatic reliability of 8–10 years, the runoff coefficient values were taken in the range of 8–10%, depending on soil characteristics and the positioning of the catchments (Doherty and Jeeves, 2015, p. 50, Tab. 2.3). By applying this method, the water runoff was calculated for basins 1–12 and basin 16.

**Method 2.** Regrarians formula – using the runoff coefficient method (Doherty, Jeeves and Pavlov, 2025):

$$Q = 100R \cdot A \cdot C \quad (2)$$

For combined surfaces with varying permeability, such as rooftops, asphalt, concrete, and gravel roads, an average surface runoff coefficient was determined based on the same source with values ranging from 0.25 to 0.90 (Doherty and Jeeves, 2015, p. 50, Tab. 2.4). By applying this method, the water runoff was calculated for basins 13–15.

**Method 3.** Formula for calculation using the specific runoff isohyet method (WMO, 1994):

$$Q = q_{sp} \cdot A \quad (3)$$

To determine the surface runoff modulus for the catchments, a raster model based on specific runoff isohyets was developed, as shown in Figure 4. From the raster model of specific runoff isohyets for the study area,  $q_{sp} = 6.04 \text{ dm}^3 \cdot \text{s}^{-1} \cdot \text{km}^{-2}$ .

The determination of runoff isohyets was carried out using annual average rainfall data from all hydrometeorological stations in Kosovo. The collected data were interpolated in GIS to generate spatial maps of runoff distribution, enabling the

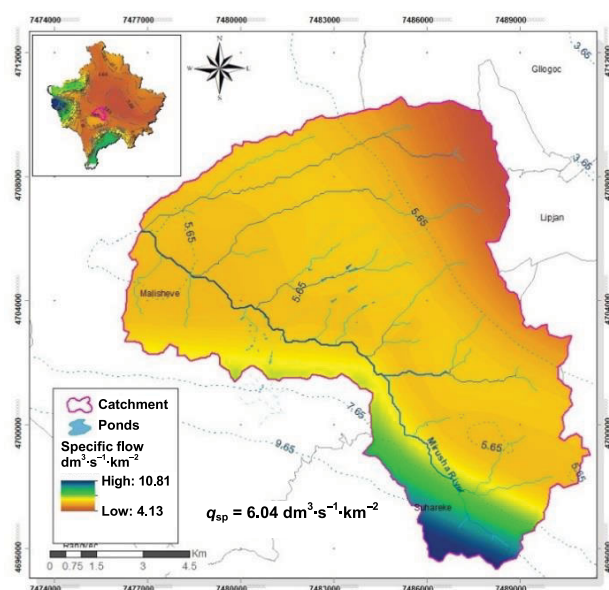


Fig. 4. Raster model of catchment areas according to specific runoff source: own elaboration based on data from IHMK (2025)

assessment of surface runoff intensity and distribution at the regional level. From these analyses, specific isohyets were derived for the study area.

## RESULTS AND DISCUSSION

Based on the principles of Keyline Design (Doherty and Jeeves, 2015), the locations of the retention ponds were determined through detailed terrain and topographic contour analysis. In each valley, a keypoint was identified – a critical point where water flow transitions from an erosive to a depositional regime and where water can be more easily retained. Along the keypoints, keylines were then drawn, defining the optimal direction for water flow and distribution across the terrain. Following this methodology, 16 retention ponds were strategically placed within the study area to maximise water capture, delay peak flows, and provide a sustainable and efficient approach for flood mitigation using non-structural methods. The locations of the 16 retention basins within the study area are illustrated in Figure 5.

Table 1 shows the annual runoff volume for the 16 retention ponds calculated using the three methods described in the section “Analysis of runoff data”. To account for hydrological variability and uncertainty in water availability, the annual runoff volume was reduced by 35% compared to the calculated value. This reduction reflects drier years, variations in runoff generation, and changes in land use in the upper catchment, such as reforestation (Worku *et al.*, 2020).

This study is based on average annual precipitation to provide a representative long-term assessment of runoff and water retention potential in the catchment. The 35% reduction scenario serves as a practical approach to represent drier years and interannual climatic variability. More detailed analyses of extreme hydrological events, such as exceptionally wet or dry years, could provide additional insights and should be considered in future studies, particularly in the context of flood modelling or climate variability scenarios.

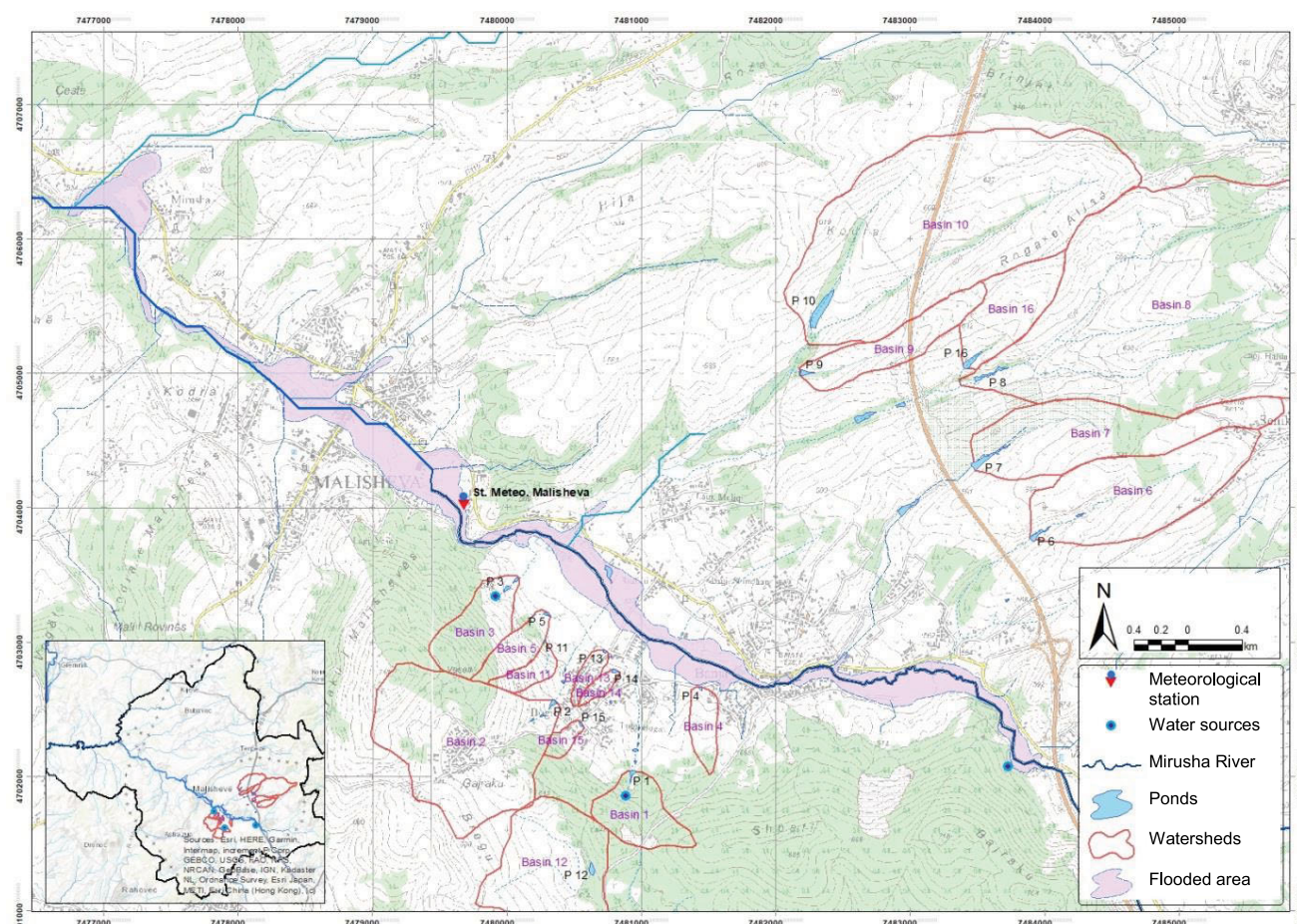


Fig. 5. Location of the retention ponds, basin and the hydrometeorological station within the study area; source: own elaboration

Table 1. Calculation of the annual runoff volume for each basin area

| Number of retention ponds | Area A (ha) | Mean annual precipitation (2006–2023) R (mm) | Regrarians formula                  |                                        |                                       |                                                                                           | Isohyetal method of specific discharge (Eq. 3)           |                                                            |                                                                           |         |
|---------------------------|-------------|----------------------------------------------|-------------------------------------|----------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------|---------|
|                           |             |                                              | runoff as % of annual precipitation | annual runoff volume                   |                                       | surface runoff module $q_{sp}$ ( $\text{dm}^3 \cdot \text{s}^{-1} \cdot \text{km}^{-2}$ ) | specific discharge ( $\text{dm}^3 \cdot \text{s}^{-1}$ ) | annual runoff volume ( $\text{m}^3 \cdot \text{yr}^{-1}$ ) | annual runoff volume reduced by 35% ( $\text{m}^3 \cdot \text{yr}^{-1}$ ) |         |
|                           |             |                                              |                                     | ( $\text{dm}^3 \cdot \text{yr}^{-1}$ ) | ( $\text{m}^3 \cdot \text{yr}^{-1}$ ) |                                                                                           |                                                          |                                                            |                                                                           |         |
| 1                         | 26.20       | 746.2                                        | Y<br>Eq. (1)                        | 8                                      | 15,637,473                            | 15,637                                                                                    | 6.04                                                     | 1.582                                                      | 49,896                                                                    | 32,432  |
| 2                         | 114.19      |                                              |                                     | 10                                     | 85,207,93                             | 85,208                                                                                    |                                                          | 6.897                                                      | 217,505                                                                   | 141,378 |
| 3                         | 28.42       |                                              |                                     | 8                                      | 16,965,329                            | 16,965                                                                                    |                                                          | 1.717                                                      | 54,133                                                                    | 35,186  |
| 4                         | 10.51       |                                              |                                     | 10                                     | 7,842,493                             | 7,842                                                                                     |                                                          | 0.635                                                      | 20,019                                                                    | 13,012  |
| 5                         | 13.44       |                                              |                                     | 8                                      | 8,024,287                             | 8,024                                                                                     |                                                          | 0.812                                                      | 25,604                                                                    | 16,642  |
| 6                         | 84.75       |                                              |                                     | 10                                     | 63,242,607                            | 63,243                                                                                    |                                                          | 5.119                                                      | 161,435                                                                   | 104,933 |
| 7                         | 81.04       |                                              |                                     | 10                                     | 60,470,129                            | 60,470                                                                                    |                                                          | 4.895                                                      | 154,358                                                                   | 100,333 |
| 8                         | 448.49      |                                              |                                     | 11                                     | 368,129,740                           | 368,130                                                                                   |                                                          | 27.089                                                     | 854,273                                                                   | 555,277 |
| 9                         | 33.02       |                                              |                                     | 10                                     | 24,639,605                            | 24,640                                                                                    |                                                          | 1.994                                                      | 62,896                                                                    | 40,882  |
| 10                        | 250.10      |                                              |                                     | 10                                     | 186,624,549                           | 186,625                                                                                   |                                                          | 15.106                                                     | 476,384                                                                   | 309,650 |
| 11                        | 8.05        |                                              |                                     | 9                                      | 5,403,433                             | 5,403                                                                                     |                                                          | 0.486                                                      | 15,326                                                                    | 9,962   |
| 12                        | 69.64       |                                              |                                     | 9                                      | 46,768,310                            | 46,768                                                                                    |                                                          | 4.206                                                      | 132,647                                                                   | 86,221  |

cont. Tab. 1

| Number of retention ponds             | Area A (ha) | Mean annual precipitation (2006–2023) <i>R</i> (mm) | Regrarians formula                  |                                      |                                     |                                                                                                    | Isohyetal method of specific discharge (Eq. 3)         |                                                          |                                                                         |           |
|---------------------------------------|-------------|-----------------------------------------------------|-------------------------------------|--------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------|-----------|
|                                       |             |                                                     | runoff as % of annual precipitation | annual runoff volume                 |                                     | surface runoff module <i>q</i> <sub>sp</sub> (dm <sup>3</sup> ·s <sup>−1</sup> ·km <sup>−2</sup> ) | specific discharge (dm <sup>3</sup> ·s <sup>−1</sup> ) | annual runoff volume (m <sup>3</sup> ·yr <sup>−1</sup> ) | annual runoff volume reduced by 35% (m <sup>3</sup> ·yr <sup>−1</sup> ) |           |
|                                       |             |                                                     |                                     | (dm <sup>3</sup> ·yr <sup>−1</sup> ) | (m <sup>3</sup> ·yr <sup>−1</sup> ) |                                                                                                    |                                                        |                                                          |                                                                         |           |
| 13                                    | 5.09        |                                                     | C<br>Eq. (2)                        | 0.75                                 | 284,696                             | 285                                                                                                |                                                        | 0.307                                                    | 9,690                                                                   | 6,298     |
| 14                                    | 3.53        |                                                     |                                     | 0.75                                 | 197,768                             | 198                                                                                                |                                                        | 0.213                                                    | 6,731                                                                   | 4,375     |
| 15                                    | 4.32        |                                                     |                                     | 0.65                                 | 209,458                             | 209                                                                                                |                                                        | 0.231                                                    | 8,226                                                                   | 5,347     |
| 16                                    | 35.92       |                                                     | Y<br>Eq. (1)                        | 8                                    | 21,442,803                          | 21,443                                                                                             | 2.170                                                  | 68,419                                                   | 44,473                                                                  |           |
| Total annual volume (m <sup>3</sup> ) |             |                                                     |                                     |                                      |                                     | 911,090                                                                                            | Total annual volume (m <sup>3</sup> )                  |                                                          | 2,317,540                                                               | 1,506,401 |

Source: own study.

Table 2 provides an illustrative example of the calculations for pond No. 7; the same approach was applied to all other ponds. This information served as the basis for the design and strategic placement of the retention ponds within the study area, allowing an accurate assessment of the maximum water storage capacity.

The results indicate that the water storage capacity of each retention pond increases progressively with rising water levels, a relationship clearly represented by the volume (Tab. 2). This table illustrates the variation of retained water volume as a function of elevation, allowing the assessment of the maximum capacity of each pond and the analysis of its hydraulic behaviour at different fill levels.

**Table 2.** Calculation of the volume for retention pond 7

| Elevation Z (m a.s.l.) | Area A ( $\text{m}^2$ ) | Cumulative area ( $\text{m}^2$ ) | Volume V ( $\text{m}^3$ ) | Cumulative volume ( $\text{m}^3$ ) |
|------------------------|-------------------------|----------------------------------|---------------------------|------------------------------------|
| 577                    | 557                     | 557                              | 278                       | 278                                |
| 578                    | 1,615                   | 2,172                            | 946                       | 1,225                              |
| 579                    | 3,370                   | 5,542                            | 2,158                     | 3,383                              |
| 580                    | 5,334                   | 10,876                           | 3,746                     | 7,129                              |
| 581                    | 7,427                   | 18,303                           | 5,586                     | 12,716                             |
| 582                    | 11,783                  | 30,086                           | 8,685                     | 21,401                             |
| 583                    | 15,791                  | 45,877                           | 12,238                    | 33,639                             |
| 584                    | 19,672                  | 65,549                           | 15,955                    | 49,593                             |

Source: own study.

Taking into account the water retention capacity of the retention ponds, the results indicate that the flooded area can be significantly reduced. Specifically, under the assumption that 60% of the runoff volume is retained by the ponds and for an average flood depth of 0.5 m, the flooded area decreases by approximately  $1.81 \text{ km}^2$  (180.77 ha). This result highlights the protective role of the retention ponds and confirms their effectiveness in reducing flood risk and managing runoff.

For the 50-year return period flood discharge (HQ50) scenario, the proposed retention ponds are capable of detaining approximately  $903,840.67 \text{ m}^3$  of floodwater. Based on this

retained volume, the reduction in the inundated area was estimated using the volume-depth relationship ( $A = V/H$ ), assuming an average flood depth of 0.50 m.

While the primary purpose of the retention ponds is to protect areas from flooding, the water stored in them can also be utilised for other practical purposes, such as agricultural irrigation. Based on the volume of water retained in each pond, the potential irrigated areas can be determined.

Considering an irrigation water requirement of  $3,750 \text{ m}^3 \cdot \text{ha}^{-1}$  and a retained water volume of  $903,840.67 \text{ m}^3$  (assuming 60% retention by the ponds), the potential irrigated area is estimated at approximately 241.02 ha.

## CONCLUSION

The strategic placement of the 16 retention ponds within the Mirusha River catchment demonstrates that non-structural measures for improving flood management can effectively reduce flooded areas and regulate surface runoff. The study shows that by accounting for hydrological variability and using conservative runoff scenarios, estimates of water retention capacity become more reliable and realistic. Additionally, the stored water can be used for complementary purposes, such as agricultural irrigation, thereby enhancing the multifunctional benefits of the retention ponds. These findings highlight the importance of integrating topographic analysis and Keyline Design principles in sustainable catchment management, while future studies should consider extreme hydrological events to further optimise flood risk management.

## ABBREVIATIONS

Q = runoff ( $\text{dm}^3 \cdot \text{s}^{-1}$  or  $\text{m}^3 \cdot \text{yr}^{-1}$ ),  
 $q_{sp}$  = surface runoff modulus ( $\text{dm}^3 \cdot \text{s}^{-1} \cdot \text{km}^{-2}$ ),  
V = volume ( $\text{m}^3$ ),  
A = catchment area (ha or  $\text{km}^2$ ),  
R = annual precipitation (mm),  
Z = elevation,  
Y = runoff as a percentage of annual precipitation,  
C = runoff coefficient,  
HQ50 = 50-year return period flood discharge.

## CONFLICT OF INTERESTS

The author declares that there are no conflicts of interest.

## INSTITUTIONAL REVIEW BOARD STATEMENT

This research does not involve animals, humans, or any other living organisms that need consent and have ethical implications.

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